Volume Day 3

Targets: G.9-4, G.9-5, G.9-3, G.9-2, G.9-1, G.9-8, G.9-7, G.9-9, G.9-10

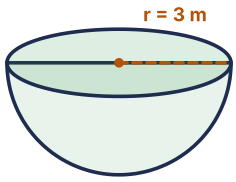
Name _____ Date _____ Block _____

Remember: Hemisphere = half a sphere: $V = \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi r^3$. **Composite solid**: find each part, then add (or subtract a hole). Missing r in a sphere: solve for r^3, then take the cube root. **Ex:** $r^3 = 27 \rightarrow r = \sqrt[3]{27} = 3$.

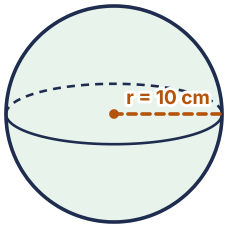
SCORE 3.0 Approaching · the basics

Targets: G.9-4, G.9-5, G.9-3, G.9-2, G.9-1

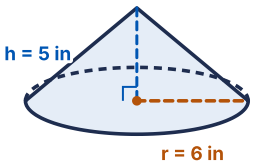
1 Find the volume of the hemisphere. Answer in terms of π. G.9-4



2 Find the volume of the sphere. Use 3.14 for π. Round to the nearest tenth. G.9-4,G.9-5

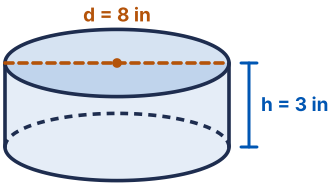


3 Find the volume of the cone. Answer in terms of π. REVIEW 9.1 G.9-3



(A) 60π in^3
(B) 180π in^3
(C) 20π in^3
(D) 10π in^3

4 Find the volume. Answer in terms of π. REVIEW 9.2 G.9-2



5 To find the radius from the diameter, multiply the diameter by 2. G.9-1

True False

SCORE 4.0 Proficient · on target

Targets: G.9-8, G.9-7, G.9-9

6 A sphere holds 36π ft^3. Find its radius. G.9-8

7 A cone holds 50π cm^3. Its radius is 5 cm. Find its height. G.9-8

8 **Ana** made a mistake. Circle it. Explain it. Fix it. G.9-7,G.9-9

ANA'S WORK

Cone ($r = 3$, $h = 8$) with a hemisphere on top

Cone: $\frac{1}{3}\pi(3)^2(8) = 24\pi$

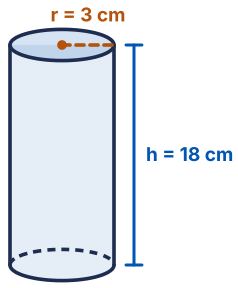
Top: $\frac{4}{3}\pi(3)^3 = 36\pi$

Total: 60π

SCORE 5.0 Mastery · go deeper

Targets: G.9-9, G.9-10

9 A can (r = 3 cm, h = 18 cm) holds 3 tennis balls. Each ball has r = 3 cm. How much space is empty? Answer in terms of π. Then round. G.9-9



Can = _____ . Balls = _____ . Empty = _____ .

10 A ball's radius is doubled. How many times more does it hold? Explain. G.9-10

It holds _____ times as much because _____ .